

Character Display and Stable Records

Keep state, presentation, transport, and evidence separate

The display shows meaning; TP14 shows transport; the record preserves facts.

Lesson	014 — component	Time	120–180 minutes
Prerequisites	Lessons 001–013	Board	Arduino Mega 2560
Evidence	D13 acquisition, LCD text, TP14 enable	Status	host verified; physical acceptance open
Safety	E1; USB-powered low-current circuit only		

Safety checkpoint. Disconnect USB before wiring. Identify the exact LCD module and its pin order from primary documentation. Do not assume its backlight includes a resistor. If the backlight current limit or polarity is unknown, leave pins A and K disconnected. Stop for heat, odor, resets, unstable power, a contrast control that places a rail across the potentiometer wiper incorrectly, or any disagreement between the identified module and the literal table. Remove USB power to stop the experiment.

1 Mission

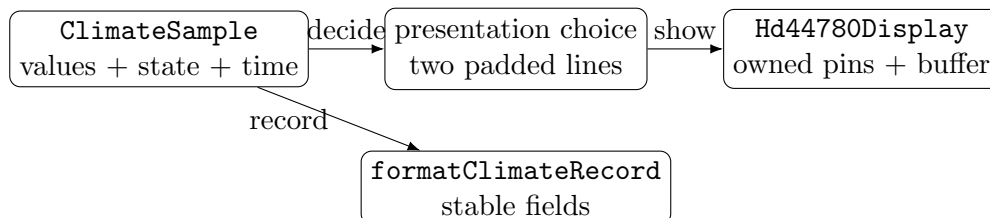
Present a deterministic climate snapshot without making the display the source of truth:

$$\text{ClimateSample} \rightarrow \begin{cases} \text{Hd44780Display} & \text{short human view} \\ \text{formatClimateRecord()} & \text{stable machine record} \end{cases}$$

The canonical sketch replays four fixed samples. This isolates presentation from sensing: the same sample selects both LCD text and a record, while neither path modifies the sample. Lesson 015 will replace the fixed table with the scheduled sensor from lesson 013.

1. Predict the first two lines and the next sequence number.
2. Observe D13 before expecting text, then observe the LCD update.
3. Observe TP14 only as transport evidence.
4. Interpret the record and display as two views of one immutable sample.

2 The four boundaries



`CharacterDisplay` is the substitutable presentation boundary. It defines lifecycle, caller-driven `update(now)`, readiness, two-row `show()`, and a desired-line snapshot. `Hd44780Display` is the concrete 16-by-2, four-bit adapter. It owns RS, E, and D4–D7 through six fixed resource claims.

Construction is inert. `initialize()` validates distinct Mega digital pins, acquires all six before exposing behavior, configures them low, and starts a staged power-up sequence. A conflict or injected configuration failure releases every claim. Repeated initialization is inert.

update(now) advances at most one startup stage or one changed character. The caller supplies time; no background task or global callback exists. The bounded one-microsecond enable pulse is an electrical transport operation, not an application schedule. **shutdown()** commands display off when possible, drives owned pins low, releases them to high impedance, and is idempotent. Destruction repeats this safely.

3 Parts and orientation

Quantity	Part	Requirement
1	Mega 2560	USB powered
1	HD44780-compatible 16-by-2 LCD	exact module and pin order identified
1	10k Ω linear potentiometer	contrast control only
1	backlight resistor	only if required by the identified module data
1	D13 indicator	built-in LED or external resistor-limited LED
1	multimeter	continuity and released-state checks
1	logic analyzer or oscilloscope	optional TP14 enable-pulse evidence

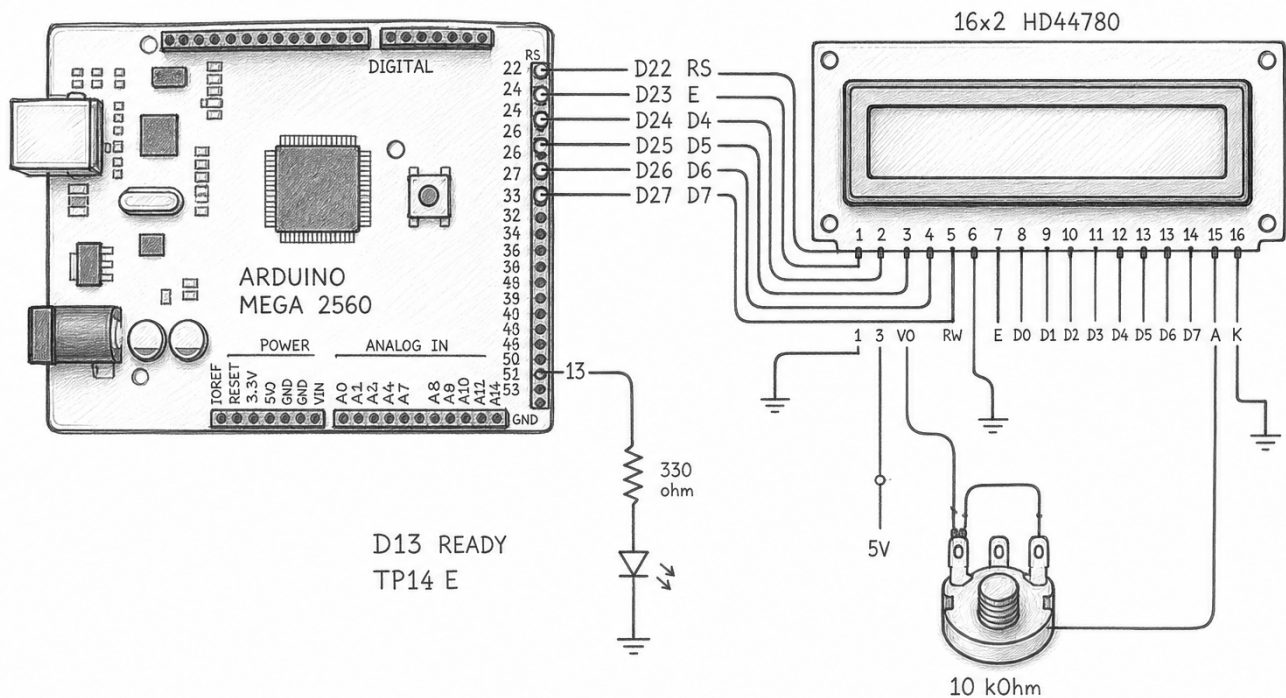


Figure 1: Pencil orientation of the Mega, LCD, contrast control, D13 evidence, and TP14. This is not an authoritative schematic; some drawn board labels are illustrative. Use the literal table and the exact module documentation. Alternative text: six Mega signal wires reach LCD RS, E, and D4 through D7; RW and VSS return to ground; VDD reaches 5 V; a potentiometer supplies VO; D13 separately indicates resource acquisition; TP14 is the E node.

4 Literal wiring

Function	Mega connection	Literal path
LCD supply	5 V and GND	identified VDD to 5 V; VSS to GND; never use Vin
contrast	VO	10 k Ω ends to 5 V/GND; wiper to identified VO
write policy	RW	identified RW directly to GND
register select	D22	D22 to identified RS
enable / TP14	D23	D23 to identified E; label this node TP14
data bit 4	D24	D24 to identified D4
data bit 5	D25	D25 to identified D5
data bit 6	D26	D26 to identified D6
data bit 7	D27	D27 to identified D7
unused data	none	D0–D3 remain unconnected in four-bit mode
backlight	A and K	connect only per identified module rating and polarity
acquisition	D13	built-in LED, or D13 through 330 Ω to LED anode

Inspect every path with USB removed. Set the contrast control near its midpoint before power. If using an external D13 LED, return its cathode to GND and record its resistor value. TP14 is measured relative to the same named GND. Place analyzer clips while unpowered.

5 Startup and incremental work

An HD44780 controller is not ready when the MCU resets. The adapter models that fact explicitly:

claim $\rightarrow$ 15 ms $\rightarrow$ 3, 3, 3, 2 nibbles $\rightarrow$ function, off, clear, entry, on $\rightarrow$ ready.

Calls made before `ready()` only advance startup when their caller time is due. Once ready, `show(row,text)` updates the desired 16-character line. It rejects a null pointer, nonprinting ASCII, an unknown row, or text longer than 16 characters without touching the LCD.

Each later `update(now)` compares desired and shown buffers and writes one changed cell. A loop therefore remains responsive while a whole line changes. The desired snapshot is stable until another `show()` call.

```
void loop ()
{
    const adk::TimePoint now (millis ());

    display.update (now);

    if (display.ready () && sampleDue (now))
    {
        const adk::ClimateSample sample = decideSample (now);
        showSample (sample);
        recordSample (sample);
    }
}
```

6 Stable records

The record format is ASCII, one line, with fixed field order and explicit units:

```
seq=42,state=valid,temp_centi_c=-125,rh_permille=456,at_ms=4294967290
```

Integers avoid decimal punctuation and floating-point formatting. Names encode units. States such as `transport.timeout`, `checksum.failure`, and `stale` remain distinct. The timestamp is the sample observation time, not the time the line happened to print.

`formatClimateRecord()` allocates nothing and does not use locale. The caller supplies a buffer. A too-small buffer is terminated and reports `CapacityExceeded`; a null or zero-capacity buffer reports `InvalidArgument`. Serial output in the canonical sketch merely carries this record. It is optional evidence: the LCD and TP14 provide the required non-Serial paths.

7 Predict, observe, interpret

Claim	Prediction and observation	Interpretation
resource acquisition	D13 turns on after all seven objects' pin claims succeed	D13 off means setup did not complete; it says nothing about LCD text
safe startup	before ready, signal pins start low and the LCD may remain blank	blank during the specified startup interval is not yet a sensor fault
transport	TP14 has bounded enable pulses as cells change	pulses prove attempted writes, not readable contrast or correct data wiring
presentation	fixed samples appear in the documented order	text order tests the presentation table and incremental flush
record identity	sequence increases once per displayed sample	one sample produced both views; locale cannot alter field punctuation
safe shutdown	D13 turns off and D22–D27 become high impedance	resource release is separate from the last visible LCD pixels

Important distinction. An LCD can retain visible pixels after MCU pins become high impedance. That persistence is not evidence that ADK still owns or drives the pins. Measure the released electrical state at a named signal pin; use the display only as semantic evidence.

8 Host experiments

Run the registered repository target after integration:

```
make host-test
make host-test-sanitize
make arduino-compile
make monitor PORT=/dev/ttyACM0 BAUD=115200
```

The deterministic tests cover:

- stopped, repeated-start, repeated-shutdown, and destructor behavior;
- staged startup from supplied timestamps;
- invalid rows, null text, nonprinting characters, and capacity limits;
- duplicate pins, resource conflict, configuration rollback, and reuse;
- one-cell incremental flushing and stable desired-line snapshots;
- exact signed values, every state name, maximum timestamps, and small record buffers;

- injected transport-write failures without hidden retries.

Run the same trace twice. Compare the transport operation sequence and record bytes. Identical inputs must produce identical outputs.

9 Bench procedure

1. **Predict.** Write the first line pair and first record below.
2. Remove USB. Identify the module pin-one mark and backlight rating.
3. Wire only supply, contrast, RW, and the six signal paths. Leave the backlight disconnected unless its current path is known.
4. Check for shorts from 5 V to GND and verify RW-to-GND continuity.
5. Place the TP14 and GND clips while unpowered.
6. Apply USB. Confirm D13 turns on before interpreting LCD behavior.
7. Adjust contrast slowly. Observe four fixed views in order.
8. Observe TP14 during changes. Do not infer timing from this drawing; record the instrument and measured pulse range.
9. Invoke or instrument shutdown. Confirm D13 off, then measure one signal pin as released rather than inferring release from the retained text.
10. Remove USB. Confirm the display and D13 are unpowered.

Predicted first view: _____

Predicted first record: _____

10 Diagnosis

Observation	Likely boundary	Next safe check
D13 remains off	claim or initialization failure	remove power; check distinct D22–D27 wiring and conflicts
D13 on, no blocks or text	supply, contrast, or startup	remove power; verify VSS/VDD/VO and module identity
blocks but no text	RS/E/data mapping	remove power; compare every literal path
wrong characters	D4–D7 order or intermittent connection	remove power; continuity-check each data path
TP14 quiet while text queued	loop halted or transport failure	inspect D13 and deterministic host trace
TP14 active, unreadable LCD	contrast, supply, or data mapping	do not treat transport as presentation proof
record differs by locale	unsupported formatter introduced	compare exact bytes to the stable grammar
old pixels after shutdown	normal LCD memory persistence	measure high impedance at a named MCU signal

11 Exercises

1. Explain why a display object owns pins while a presentation table owns no hardware.

2. Add a fourth record consumer that counts states without parsing LCD text. What boundary should it accept?
3. Calculate the maximum number of loop calls needed to replace both 16-character lines when one dirty cell flushes per call.
4. Add a test for -32768 centi-degrees and explain why negation first widens to 32 bits.
5. Design a 20-by-4 extension without changing the climate sample or stable-record grammar. Which dimensions become configuration?
6. Inject a write failure on the enable-low operation. Predict the status, desired snapshot, and caller's safe response.
7. Explain why LCD pixel persistence cannot prove output shutdown.

12 Acceptance record

Item	Record
Board and USB supply	_____
LCD manufacturer/module/revision	_____
Primary pinout source	_____
Backlight rating/path, or disconnected	_____
Contrast potentiometer measured value	_____
D13 evidence behavior	_____
TP14 instrument and pulse observation	_____
Four displayed views in order	_____
Stable record bytes checked	_____
Shutdown pin and measured released state	_____
Reset and power-removal observations	_____
Deviation, reviewer, date	_____

Current release statement: implementation and deterministic host tests are complete for this package. Mega compilation and size evidence are recorded by the integrator. Physical behavior, electrical levels, contrast, backlight current, timing, shutdown, reset, and power removal remain **open** until the table above contains measured results.

13 Next

Lesson 015 composes the scheduled climate sensor, this character display, stable records, and explicit health policy into an environmental station. It does not turn this learning circuit into a calibrated monitor or safety alarm.